

Faculty of Engineering and Technology

Department of Electrical and Computer Engineering

ENCS3310

ADVANCED DIGITAL DESIGN

PROJECT

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Abstract

This project presents the design and verification of a smart traffic light controller with a pedestrian request using a finite state machine (FSM). The controller manages traffic lights for a main road, a side road, and a pedestrian crossing, where the main road is green by default. State transitions occur when a vehicle is detected on the side road or when the pedestrian button is pressed.

The system follows fixed timing requirements for green states and supports queuing pedestrian requests during side-road operation. The design is implemented in Verilog and verified using a self-checking testbench. Simulation results confirm correct behavior according to the given specifications.

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FSM state diagram

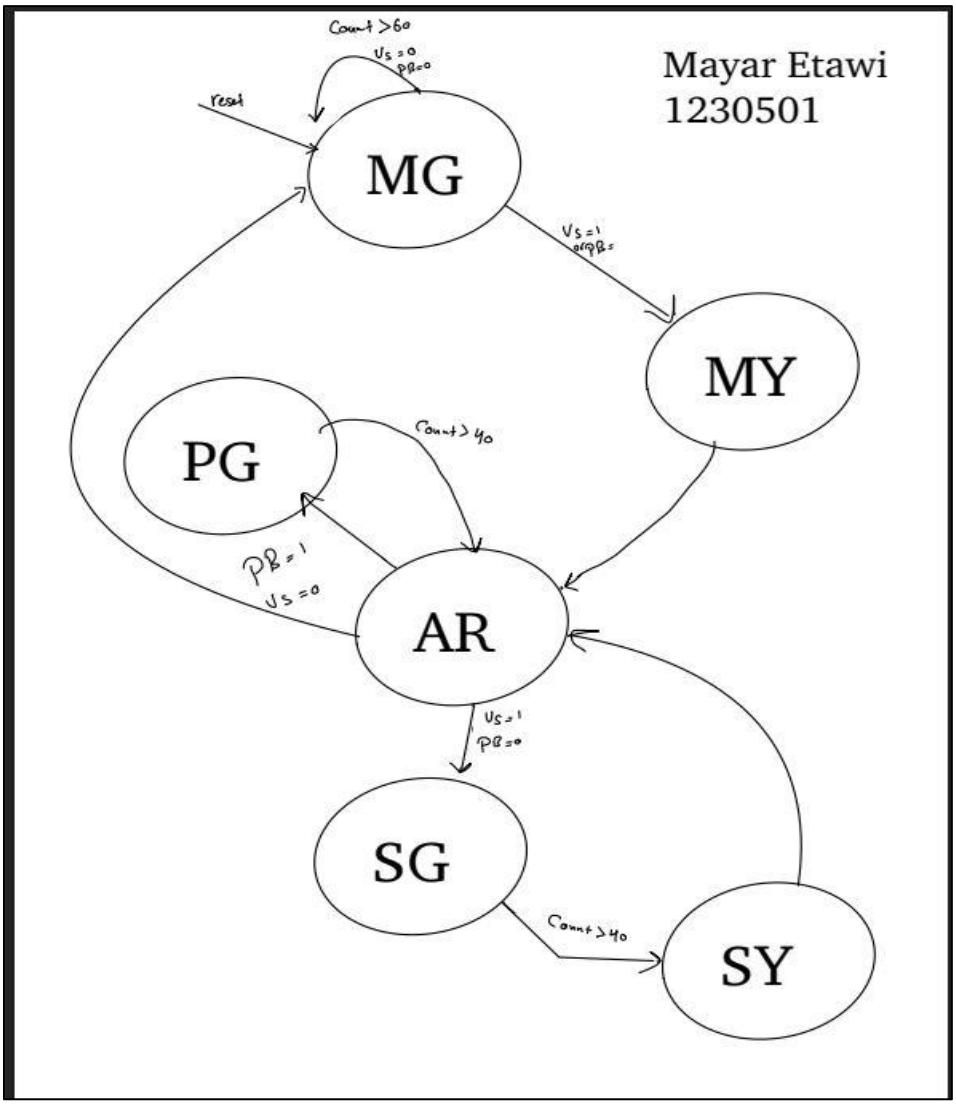


Figure 1: state diagram.

The state diagram illustrates the operation of the traffic light controller using a finite state machine (FSM). The system starts in the default state MG (Main Green), where the main road has a green light and the side road is red after reset. This transition represents the default operation from red to green on the main road. When a vehicle sensor (VS) or pedestrian button (PB) request occurs and the minimum green time is satisfied, the controller transitions through the yellow state and an all-red (AR) state to ensure safety. From the all-red state, the system moves either to the side road green state (SG) or the pedestrian green state (PG), giving priority to pedestrian requests. This structured sequence guarantees safe and organized traffic flow.

State Encoding

The traffic light controller is designed using a Finite State Machine (FSM) approach, where binary encoding is used to represent the different system states. The system consists of six main states that correspond to the operating phases of the traffic lights. These states include Main Green (MG), Main Yellow (MY), All Red (AR), Side Green (SG), Side Yellow (SY), and Pedestrian Green (PG).

Since the total number of states in the system is six, a 3-bit binary encoding is selected, as at least three bits are required to uniquely represent six different states. This choice ensures that each state has a unique binary value, preventing ambiguity during state transitions while maintaining a simple and efficient hardware design.

In addition to state encoding, a 3-bit encoding scheme is also used to represent the traffic light colors (RED, YELLOW, and GREEN) for both the main road and the side road. This encoding allows each light color to be clearly and independently represented at the output. Using 3-bit encoding for both states and light colors simplifies the overall design, improves readability during simulation and waveform analysis, and ensures consistency with the project specifications.

Timing Behavior

The operation of the traffic light controller is governed by timing constraints that control the duration of each green state. A counter is used within the FSM to track the time spent in the current state and to determine when a transition to the next state should occur. The main road green state (MG) has a minimum duration of 60 clock cycles, during which the controller remains in this state as long as no vehicle is detected on the side road and no pedestrian request is issued.

The side road green state (SG) and the pedestrian green state (PG) have fixed durations of 40 clock cycles each. Once these durations expire, the controller transitions to the next state according to the FSM logic. Transitional states such as the yellow and all-red states are used to ensure safe switching between traffic flows and are executed sequentially without relying on additional timing counters.

The timing mechanism is reset whenever a state transition occurs, ensuring accurate measurement of each state’s duration. This approach guarantees that all timing requirements specified in the project are satisfied while maintaining a clear and deterministic state transition behavior.

Results

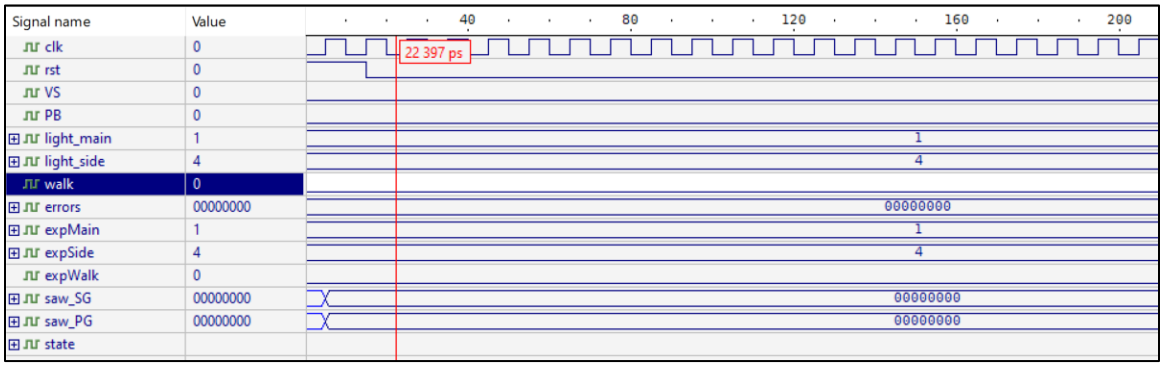


Figure 2: Default (VS = 0, PB = 0)

The simulation results for this case demonstrate the default behavior of the traffic light controller when there are no requests from either the side-road vehicle sensor or the pedestrian button. After the reset signal is deasserted (rst = 0) and the clock stabilizes, the system remains in the **Main Green (MG)** state. In this state, the main road traffic light is green (light_main = GREEN), the side road traffic light is red (light_side = RED), and the pedestrian walk signal remains inactive (walk = 0).

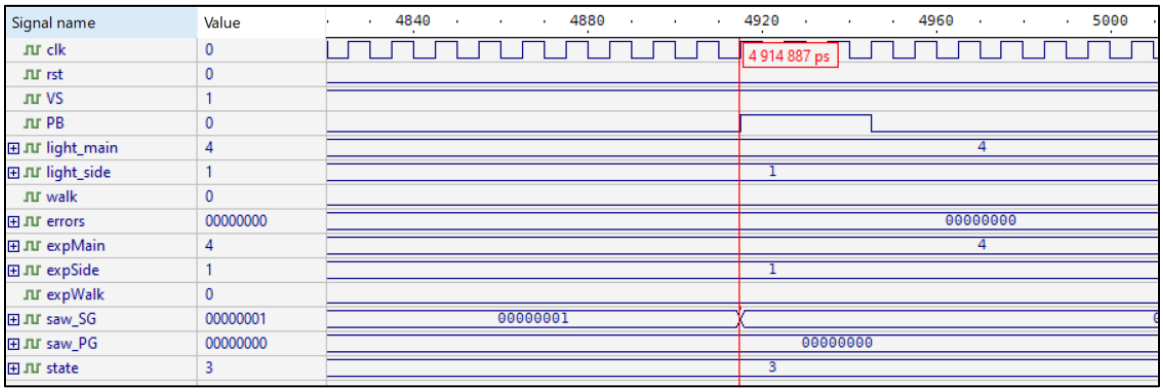


Figure 3: Side-Road Vehicle Request (VS = 1, PB = 0)

In this case, the vehicle sensor on the side road is activated while no pedestrian request is present. As shown in the waveform, after the minimum main green time expires, the controller transitions from **Main Green (MG)** through the intermediate states and reaches the **Side Green (SG)** state. During this state, the side road traffic light turns green (light_side = GREEN), while the main road traffic light turns red (light_main = RED), and the pedestrian walk signal remains disabled (walk = 0).

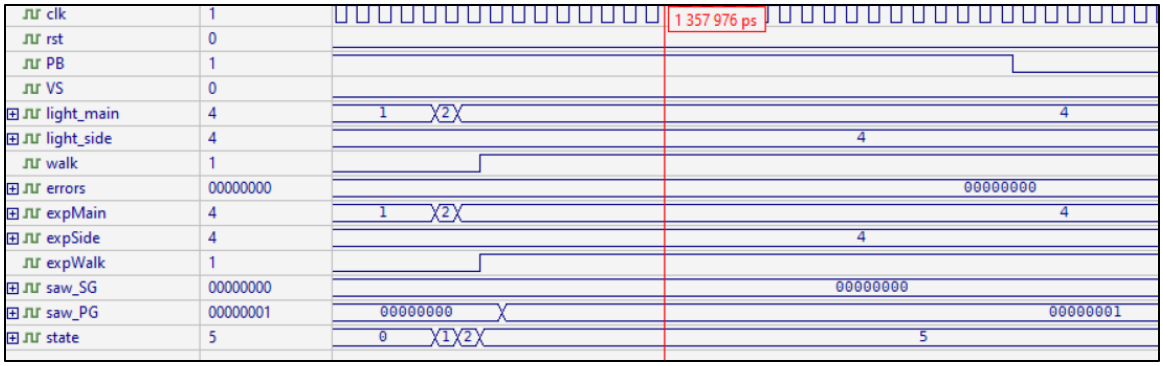


Figure 4: Pedestrian Request (PB = 1, VS = 0).

When the pedestrian button (PB) is asserted while no vehicle request is present on the side road (VS = 0), the controller completes the minimum Main-Green duration, then transitions through Main-Yellow and All-Red states before entering the Pedestrian-Green (PG) state. During the PG state, the walk signal is asserted, allowing pedestrians to cross safely while both traffic lights remain red.

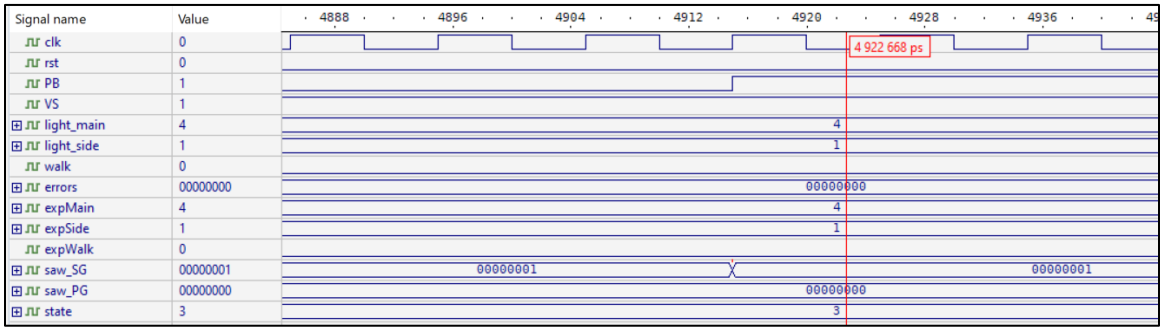


Figure 5: Request (PB = 1, VS =1).

When both the vehicle sensor (VS) and pedestrian button (PB) are asserted simultaneously, the controller gives priority to the pedestrian request. After completing the minimum main green duration and passing through the yellow and all-red states, the FSM transitions to the pedestrian green (PG) state. This ensures safe pedestrian crossing while both traffic directions are stopped.

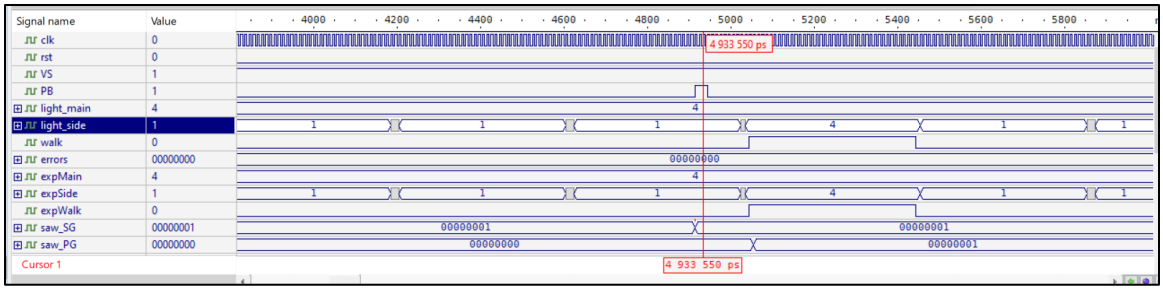


Figure 6: Pedestrian Request During Side-Green (Queued Request).

In this case, a pedestrian request (PB = 1) is asserted while the system is in the Side-Green (SG) state. As shown in the waveform, the walk signal remains LOW during SG, indicating that the pedestrian request is not served immediately. Instead, the request is stored (queued) internally. After the Side-Green duration completes and the controller transitions through the intermediate states, the queued pedestrian request is served by entering the Pedestrian-Green (PG) state.

Conclusion

In this project, a smart traffic light controller with pedestrian support was successfully designed and implemented using a finite state machine (FSM). The system correctly handles vehicle and pedestrian requests while respecting the specified timing constraints. Simulation results and waveform analysis confirm that all defined scenarios behave as expected, including request prioritization and queuing. The self-checking testbench verified the correctness of the design against the specification, demonstrating a reliable and well-structured RTL implementation.